Talk on Bayesian Optimization

Jungtaek Kim
(jtkim@postech.ac.kr)

Machine Learning Group,
Department of Computer Science and Engineering, POSTECH,
77-Cheongam-ro, Nam-gu, Pohang-si 37673,
Gyungsangbuk-do, Republic of Korea

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Bayesian Optimization

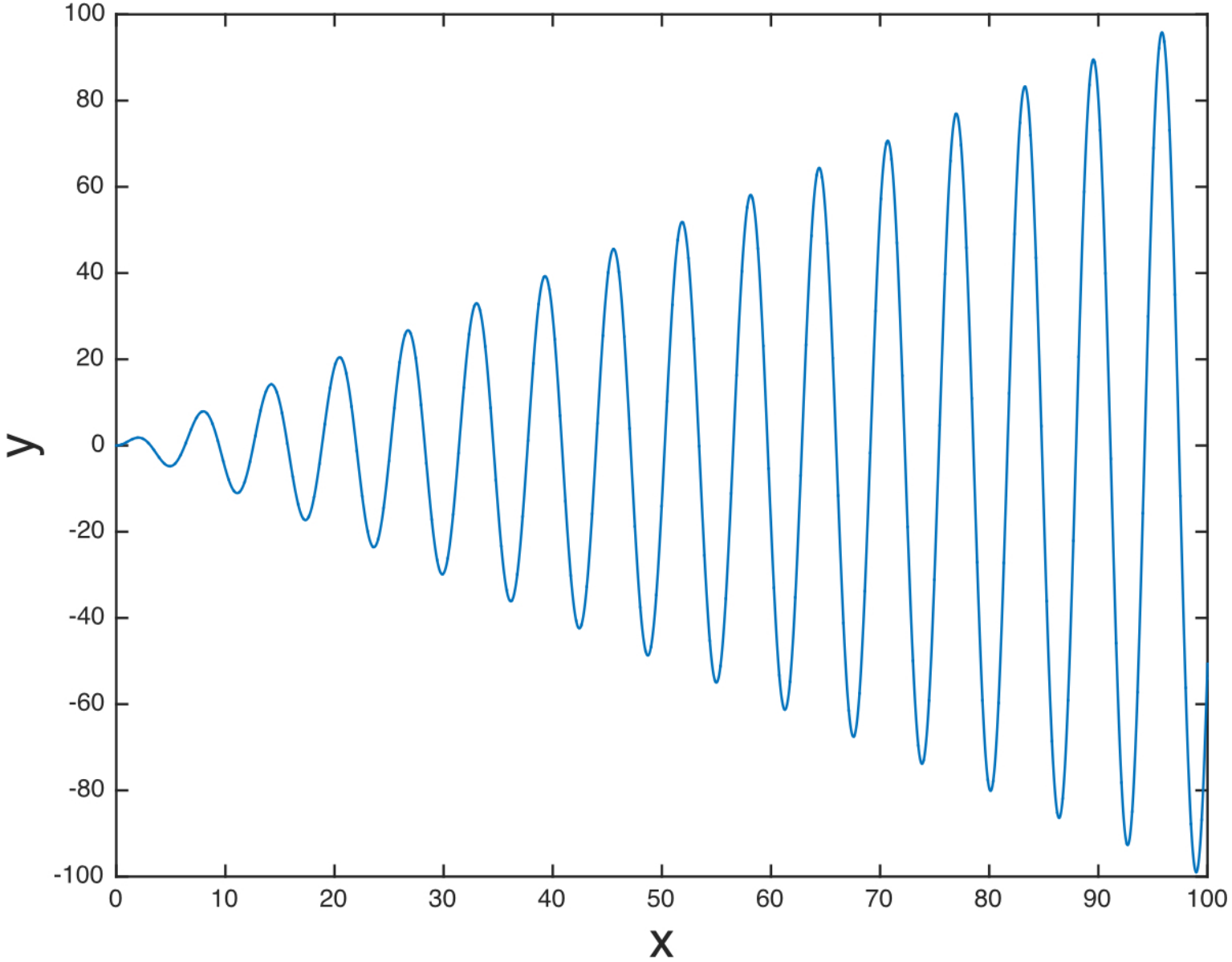
Bayesian Optimization for Expensive Black-box Function

- A powerful strategy for finding the extrema of objective functions that are expensive to evaluate,
  - where one does not have a closed-form expression for the objective function, but where one can obtain observations at sampled values.
- The prior represents our belief about the space of possible objective functions.
- The posterior distribution is

$$P(f|\mathcal{D}_{1:t}) \propto P(\mathcal{D}_{1:t}|f)P(f),$$

where $\mathcal{D}_{1:t} = \{\mathbf{x}_{1:t}, f(\mathbf{x}_{1:t})\}$, $f(\mathbf{x}_i)$ is the observation of the objective function at $\mathbf{x}_i$, $P(f)$ is the prior distribution, and $P(\mathcal{D}_{1:t}|f)$ is the likelihood.

[Brochu, et al., 2009]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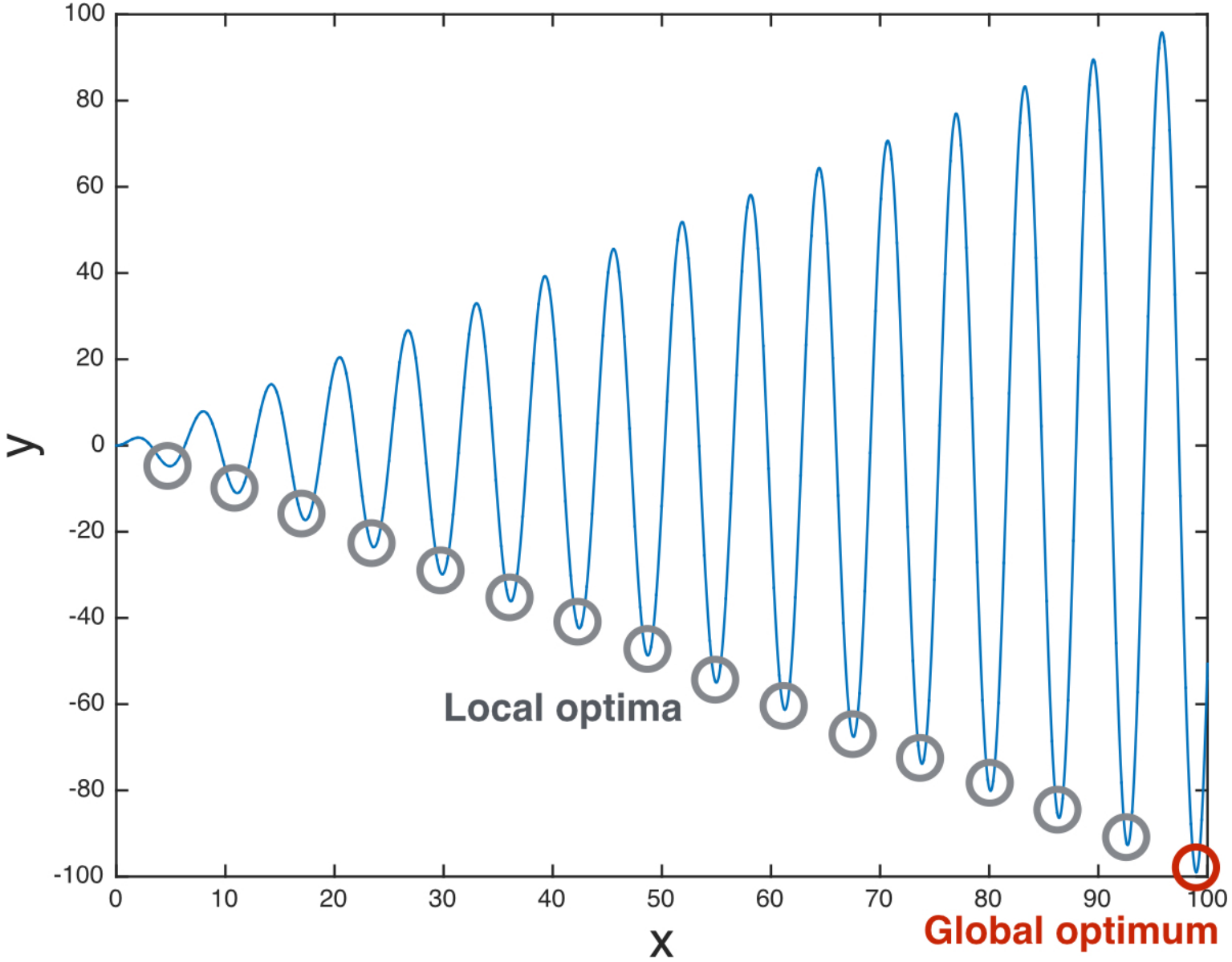
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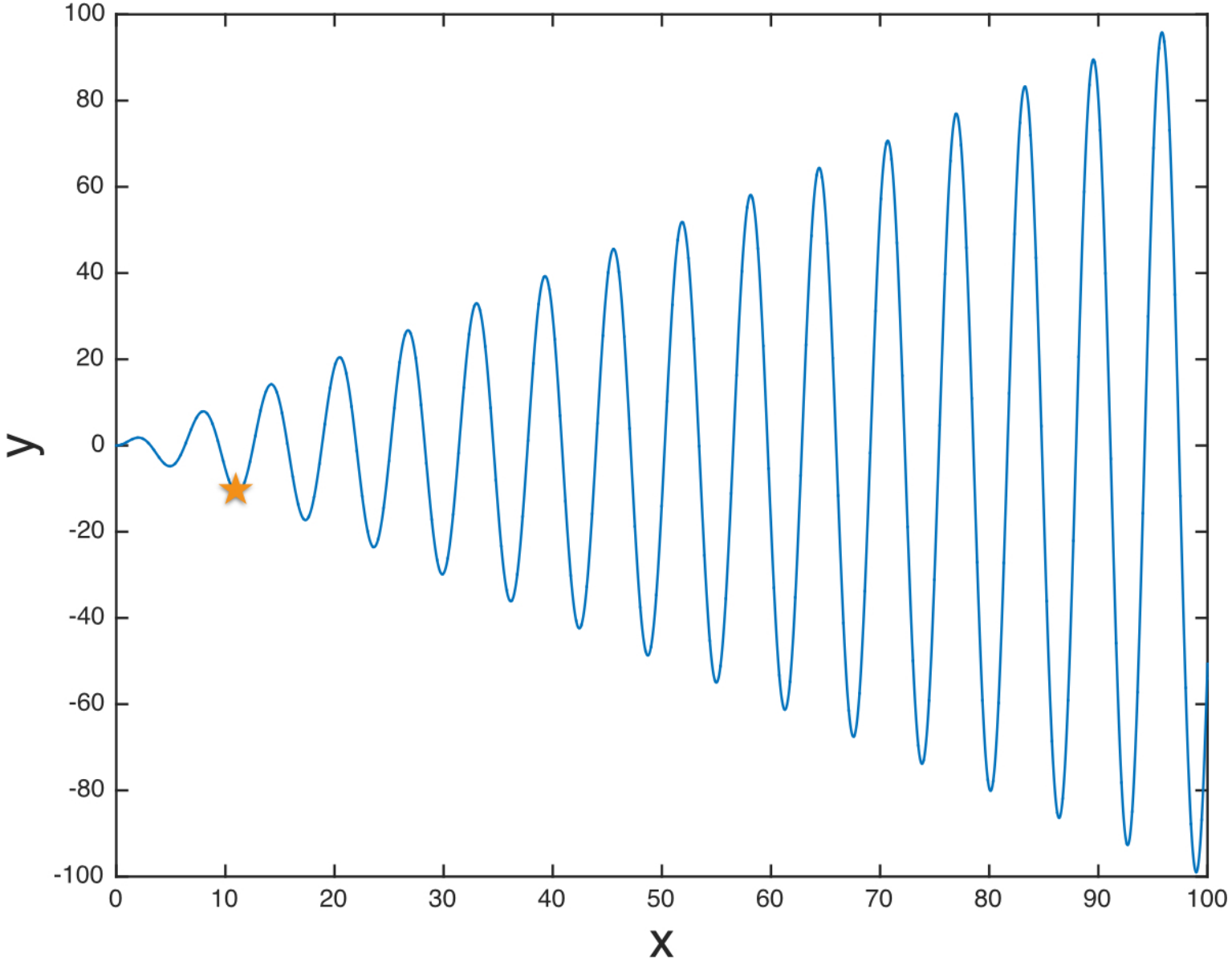
Global optimum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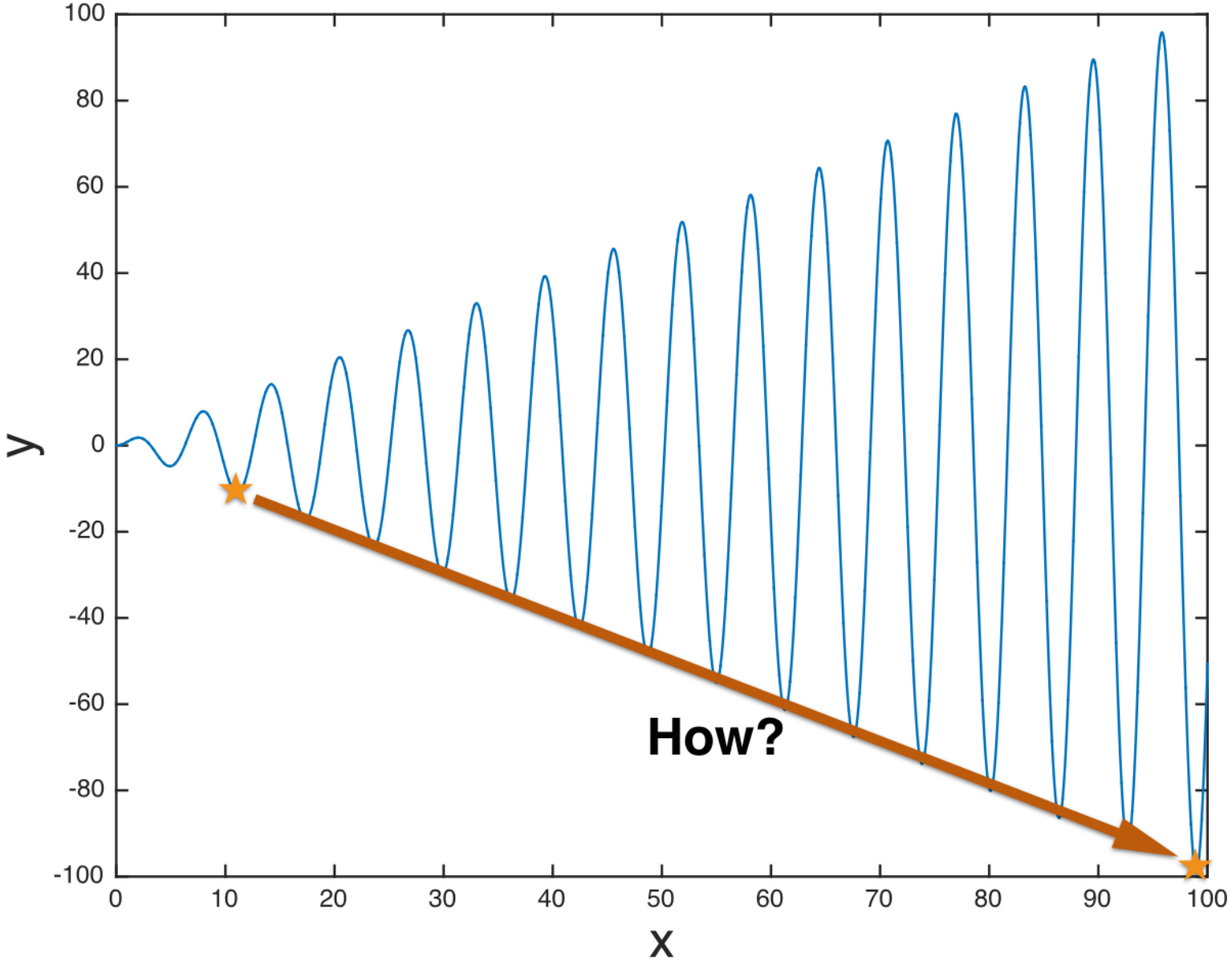
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Algorithm of Bayesian Optimization

Algorithm 1 Bayesian Optimization

Require: Initial data $\mathcal{D}_{1:I} = \{(\mathbf{x}_i, y_i)_{1:I}\}$.

1: **for** $t = 1, 2, \ldots,$ **do**

2: Predict a function $f^*(\mathbf{x}|\mathcal{D}_{1:I+t-1})$ considered as an objective function.

3: Find $\mathbf{x}_{I+t}$ by optimizing the acquisition function, $\mathbf{x}_{I+t} = \arg\max_{\mathbf{x}_{I+t}} a(\mathbf{x}|\mathcal{D}_{1:I+t-1})$.

4: Sample the objective function, $y_{I+t} = f(\mathbf{x}_{I+t}) + \epsilon_{I+t}$.

5: Update on $\mathcal{D}_{1:I+t} = \{\mathcal{D}_{1:I+t-1}, (\mathbf{x}_t, y_t)\}$.

6: **end for**

Supplement: Gaussian Process

- A Gaussian Process (GP) is a collection of random variables, any finite number of which have a joint Gaussian distribution.
- Generally, GP is expressed as

$$f \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$$

where

$$\begin{aligned} m(\mathbf{x}) &= \mathbb{E}[f(\mathbf{x})] \\ k(\mathbf{x}, \mathbf{x}') &= \mathbb{E}[(f(\mathbf{x}) - m(\mathbf{x}))(f(\mathbf{x}') - m(\mathbf{x}'))]. \end{aligned}$$

[Rasmussen and Williams, 2006]

Supplement: Gaussian Process Regression

- The squared-exponential covariance functions in one dimension has the following form,

$$k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp(-\frac{1}{2l^2}(\mathbf{x} - \mathbf{x}')^2) + \sigma_n^2 \delta_{\mathbf{x}\mathbf{x}'},$$

  where σ_f is the signal standard deviation, l is the length scale and σ_n is the noise standard deviation.
- The mean and covariance of the predictive distribution are

$$\begin{aligned} Mean &= K(X_{test}, X_{training})(K(X_{training}, X_{training}) + \sigma_n^2 I)^{-1} y, \\ Covariance &= K(X_{test}, X_{test}) \\ &\quad - K(X_{test}, X_{training})(K(X_{training}, X_{training}) + \sigma_n^2 I)^{-1} K(X_{training}, X_{test}). \end{aligned}$$

[Rasmussen and Williams, 2006]

Acquisition Functions for Bayesian Optimization

- Acquisition function is a function that acquires a next point to evaluate for a black-box expensive function.
- Traditionally, the Probability of Improvement (PI) [Kushner, 1964], the Expected Improvement (EI) [Mockus et al., 1978], and GP-Upper Confidence Bound (GP-UCB) [N. Srinivas et al., 2010] are used for Bayesian optimization.
- Several functions like Predictive Entropy Search (PES) [Hernandez-Lobato et al., 2014] and a combination of existing functions are suggested recently.

Traditional Acquisition Functions

- PI

$$a_{PI}(\mathbf{x};\{\mathbf{x}_n,y_n\},\theta)=\Phi(Z),$$

- EI

$$a_{EI}(\mathbf{x};\{\mathbf{x}_n,y_n\},\theta)=\begin{cases}(\mu(\mathbf{x})-f(\mathbf{x}^+))\Phi(Z)+\sigma(\mathbf{x})\phi(Z) & \text{if } \sigma(\mathbf{x})>0\\ 0 & \text{if } \sigma(\mathbf{x})=0\end{cases},$$

- GP-UCB

$$a_{UCB}(\mathbf{x};\{\mathbf{x}_n,y_n\},\theta)=\mu(\mathbf{x},\{\mathbf{x}_n,y_n\},\theta)+\beta\sigma(\mathbf{x};\{\mathbf{x}_n,y_n\},\theta),$$

where

$$Z=\begin{cases}\frac{\mu(\mathbf{x})-f(\mathbf{x}^+)}{\sigma(\mathbf{x})} & \text{if } \sigma(\mathbf{x})>0\\ 0 & \text{if } \sigma(\mathbf{x})=0\end{cases}.$$

Probability of Improvement

- PI is given with a trade-off parameter $\xi \geq 0$,

$$
\begin{aligned}
a_{PI}(\mathbf{x}) &= \mathbb{E}(\mathbb{I}^0) \\
&= P(\mu(\mathbf{x}) \geq f(\mathbf{x}^+) + \xi) \\
&= \Phi(\frac{\mu(\mathbf{x}) - f(\mathbf{x}^+) - \xi}{\sigma(\mathbf{x})}),
\end{aligned}
$$

where $\mathbb{I} = \max\{0, \mu(\mathbf{x}) - f(x^+) - \xi\}$.

Example of PI

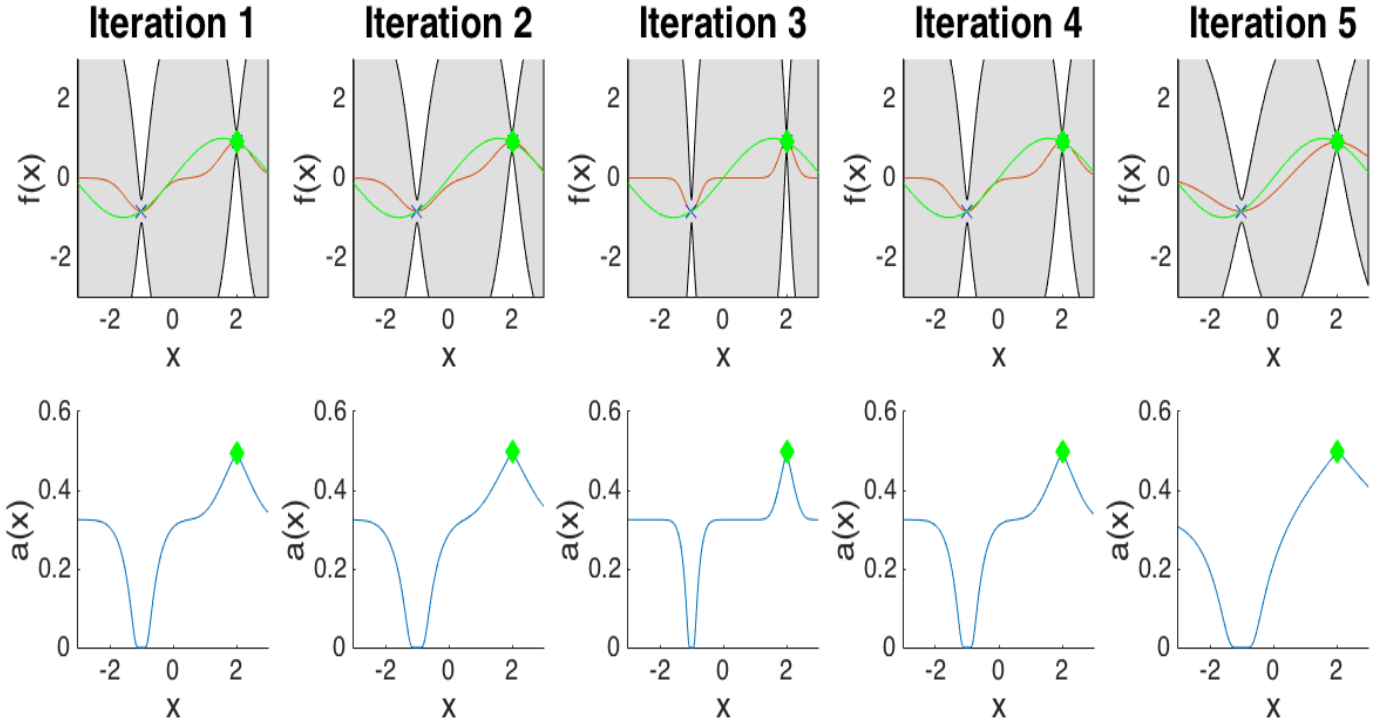
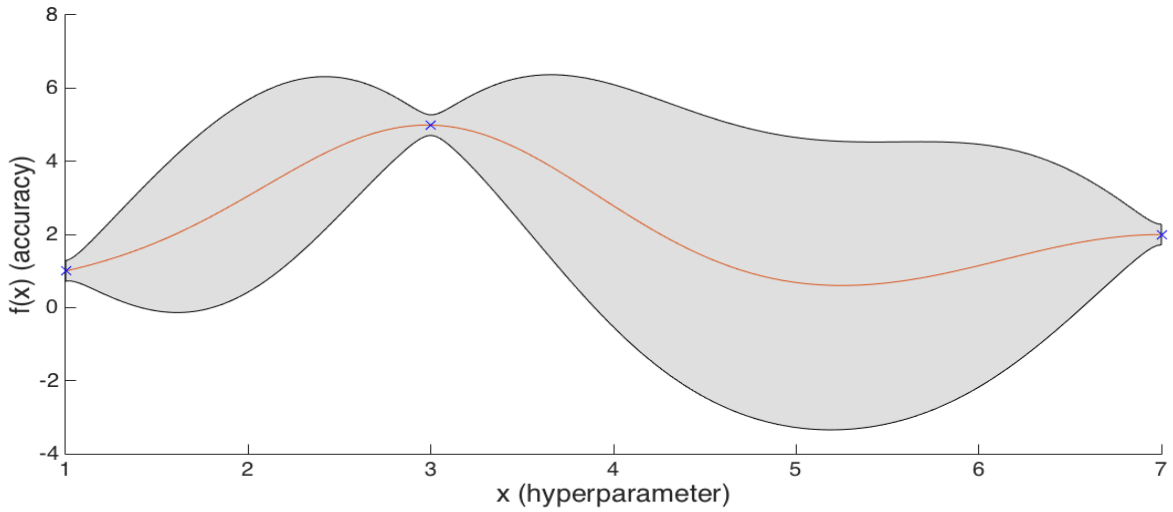
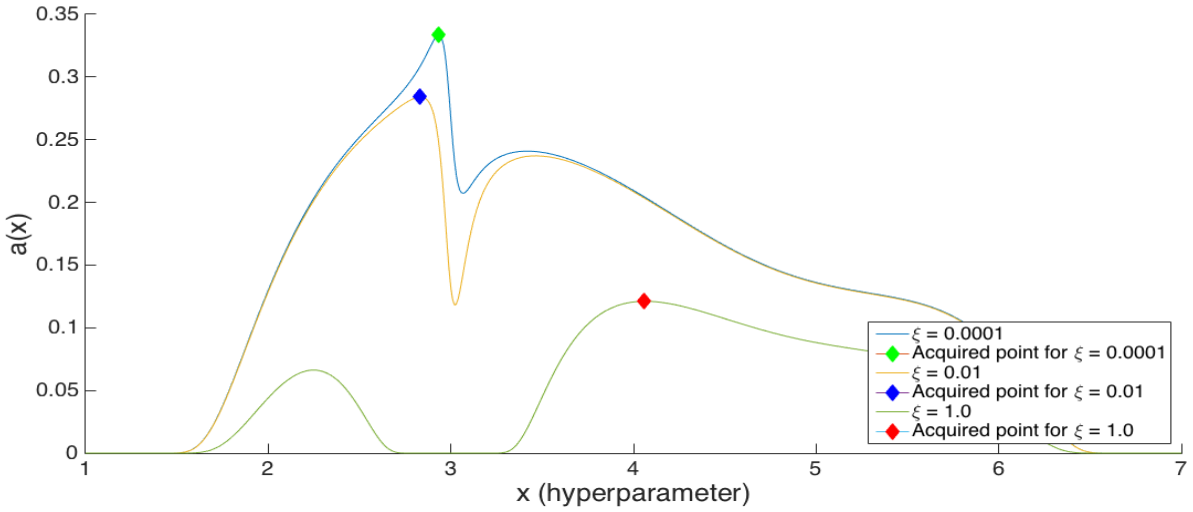


Figure 1: Objective function is red and acquisition function is blue. Green point is the last acquired point and x point is a training data.

Exploration-exploitation trade-off of PI





Expected Improvement

- EI is given with a trade-off parameter $\xi \geq 0$,

$$a_{EI}(\mathbf{x}) = \mathbb{E}(\mathbb{I}^1)$$
$$= \begin{cases} (\mu(\mathbf{x}) - f(\mathbf{x}^+) - \xi)\Phi(Z) + \sigma(\mathbf{x})\phi(Z) & \text{if } \sigma(\mathbf{x}) > 0 \\ 0 & \text{if } \sigma(\mathbf{x}) = 0 \end{cases},$$

where

$$\mathbb{I} = \max\{0, \mu(\mathbf{x}) - f(x^+) - \xi\}$$

and

$$Z = \begin{cases} \frac{\mu(\mathbf{x}) - f(\mathbf{x}^+) - \xi}{\sigma(\mathbf{x})} & \text{if } \sigma(\mathbf{x}) > 0 \\ 0 & \text{if } \sigma(\mathbf{x}) = 0 \end{cases}.$$

Example of EI

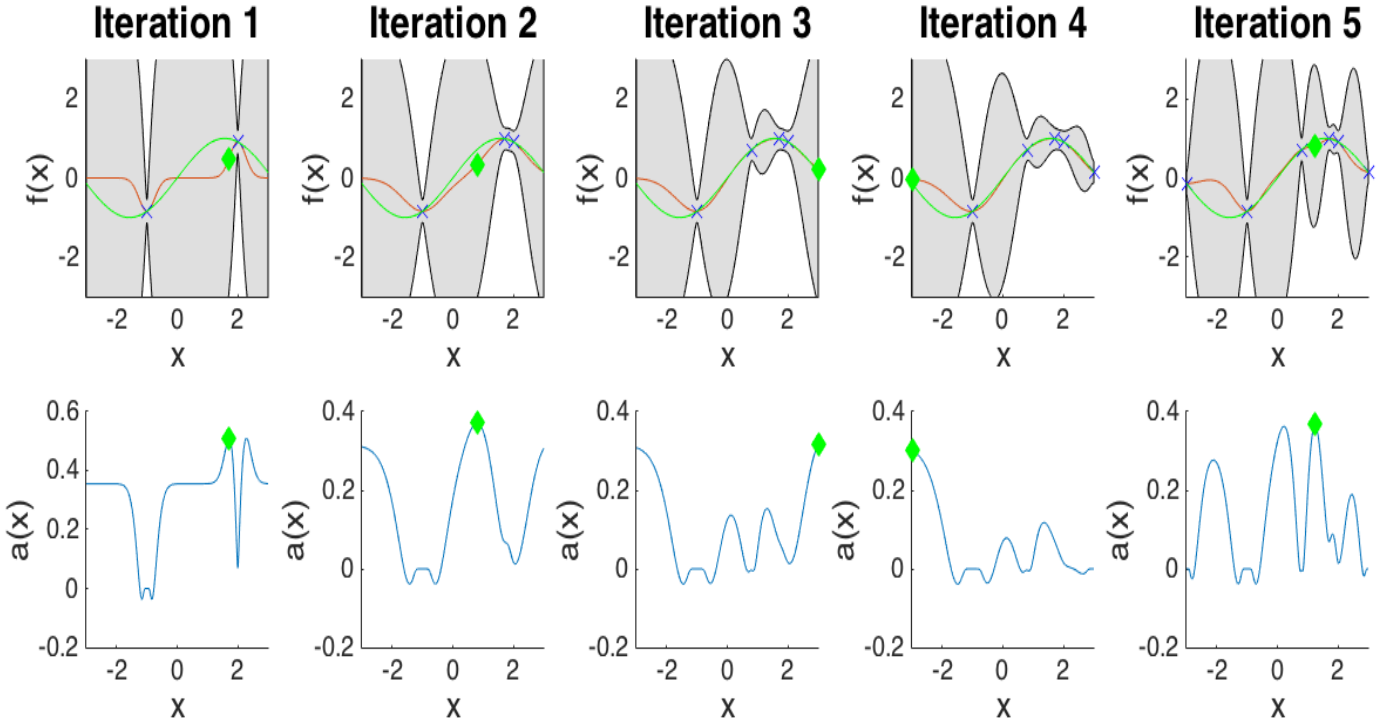
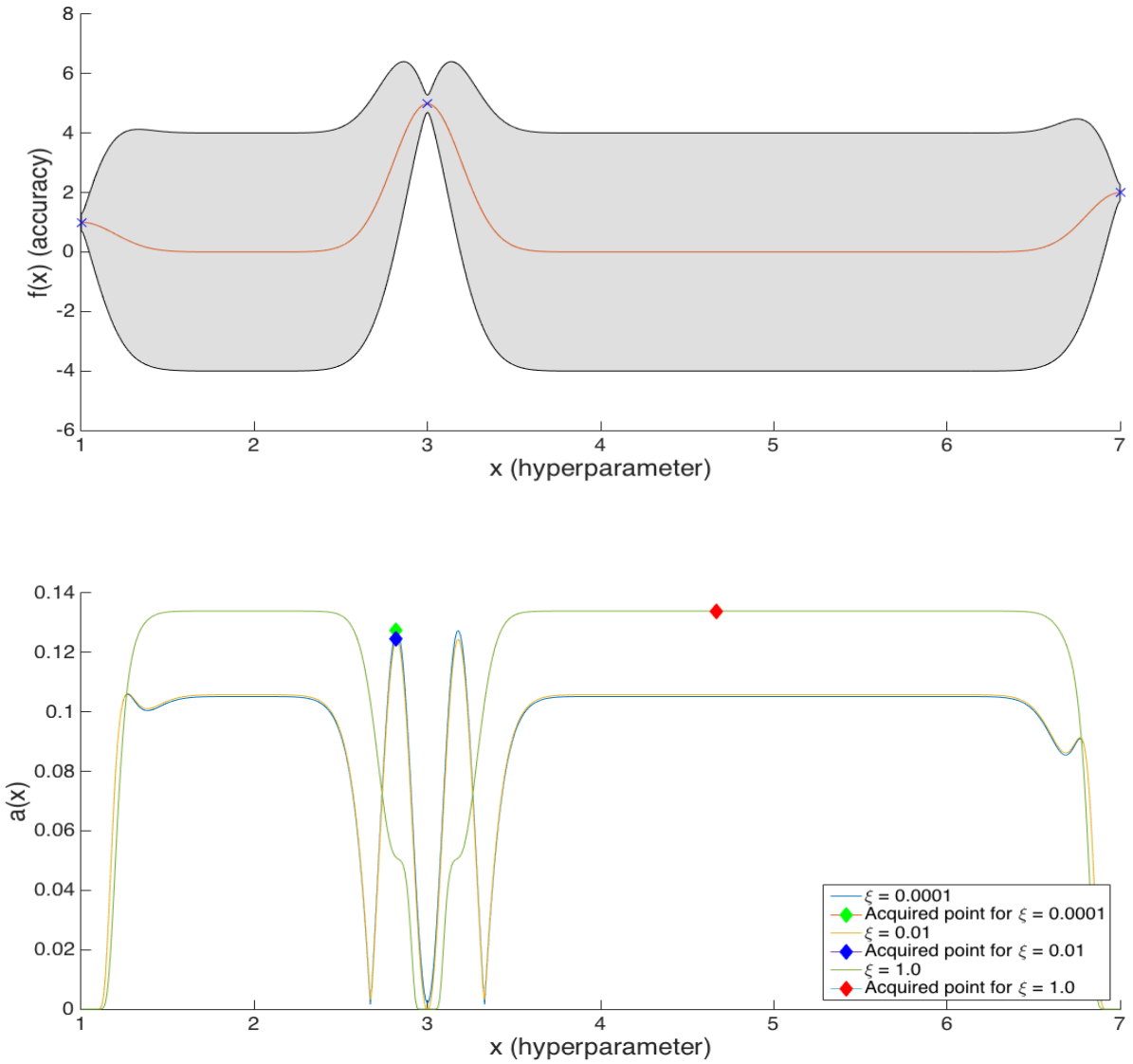


Figure 2: Objective function is red and acquisition function is blue. Green point is the last acquired point and x point is a training data.

Exploration-exploitation trade-off of EI



GP-Upper Confidence Bound

- GP-Upper Confidence Bound is

$$a_{UCB}(\mathbf{x}) = \mu(\mathbf{x}) + \beta\sigma(\mathbf{x}),$$

where β is given.

Example of GP-UCB

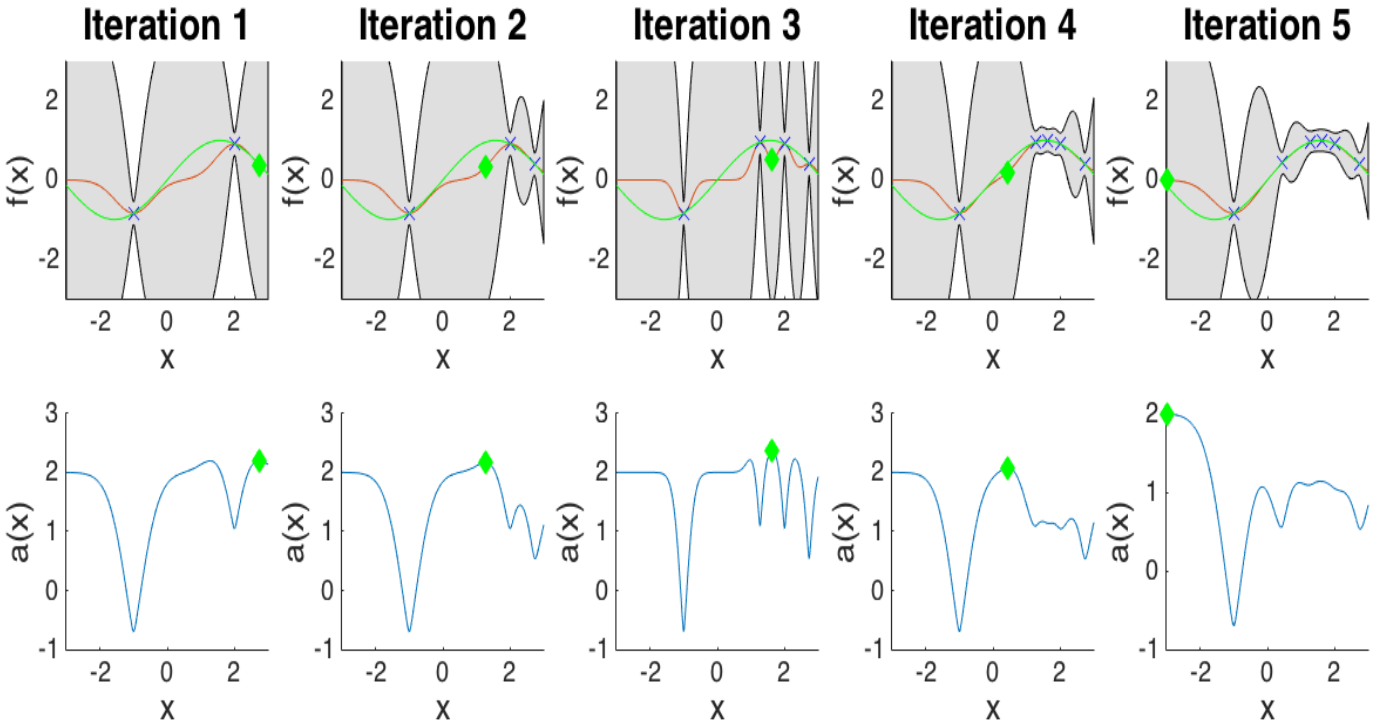
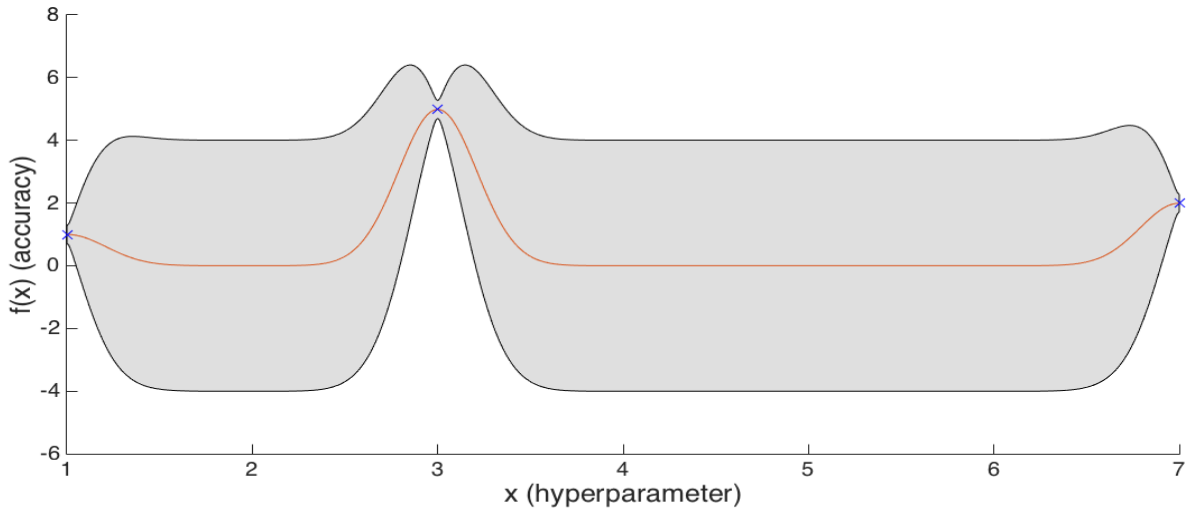
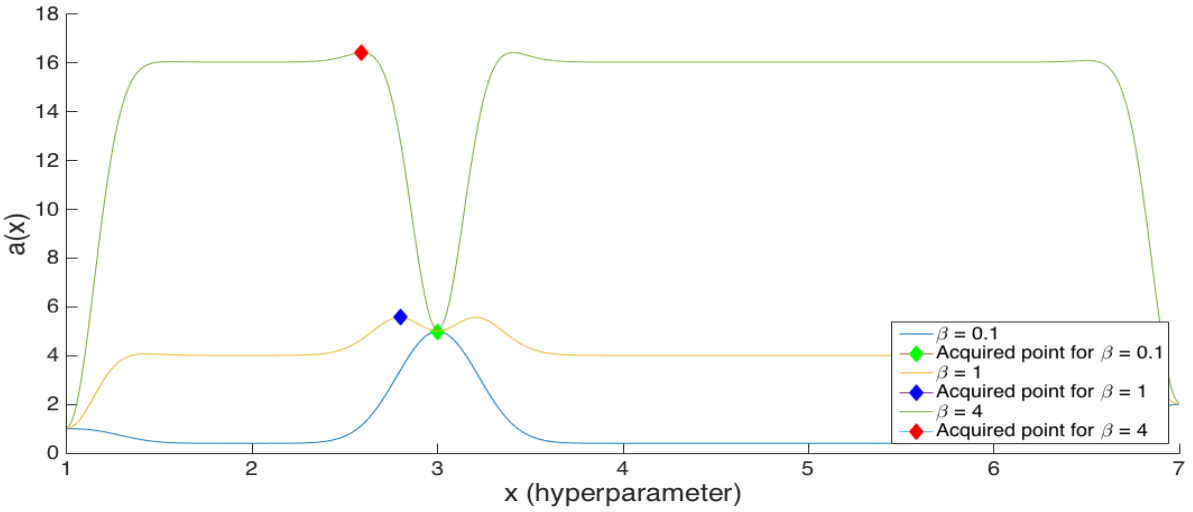


Figure 3: Objective function is red and acquisition function is blue. Green point is the last acquired point and x point is a training data. β is 1.0.

Exploration-exploitation trade-off of GP-UCB





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